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Automating Core Data Integration: Insights from the Volve Field Case Study

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Abstract

This study presents an integrated, multi-scale workflow for quality evaluation in Middle to Late Jurassic reservoir in the Volve Field, Norwegian North Sea (Hugin Formation), focusing on Volve 15/9-19A and 15/9-19BT2 wells. It aims to streamline core data interpretation from reservoir to pore scale using machine learning algorithms, enhance reservoir characterization efficiency and accuracy through the unification of geological and petrophysical data across different spatial resolutions.

The proposed workflow integrates core gamma-ray responses with high-resolution core image analysis and applies a random forest classifier to automate lithological descriptions. These outputs are refined through integration with thin-section petrography for detailed facies analysis. Facies are then correlated with core petrophysical measurements and poral network characteristics derived from microscopy. Finally, K-means clustering is applied to petrophysical well logs into discrete reservoir quality classes. The integration of these classes with facies analysis and petrophysical evaluation enabling a robust, data-driven approach to reservoir zonation and quality assessment.

Automated lithological description significantly reduces interpretation time while maintaining consistency and quality. Multi-scale reservoir characterization within the Hugin Formation enabled identification of three distinct reservoir quality zones in the Volve 15/9-19A well, and two zones in the 15/9-19BT2 well. These zones are separated by lithological and diagenetic heterogeneities.

Reservoir zone one (23.74 m thick) on the Volve 15/9-19A well, is dominated by mouth bar, tidal bar, and upper shoreface facies. This interval is primarily associated with good reservoir quality, with localized sections of excellent quality. However, the presence of lithological heterogeneities results in localized poor to moderate reservoir quality. Reservoir zone two (19.32 m thick) is dominated by tidal channel facies, with minor contribution from tidal bar facies. This zone is largely associated with excellent reservoir quality, with subordinate intervals of good quality. Reservoir zone three (12.96 m thick), is composed mainly of mouth bar and tidal channel facies. It is predominantly characterized by good reservoir quality, with minor sections exhibiting excellent quality. However, the presence of lithological heterogeneities (represented by tidal flat facies) and diagenetic overprint within the mouth bar facies leads to localized poor reservoir quality.

In the Volve 15/9-19BT2 well, two reservoir quality zones were identified within the Hugin Formation. Reservoir zone one (17.43 m thick) is dominated by tidal channel facies. This interval is primarily

characterized by good reservoir quality, with minor sections exhibiting moderate quality. Limited diagenetic heterogeneities are observed near the top of the zone, locally reducing reservoir quality. Reservoir zone two (26.70 m thick), comprises a mix of tidal bar and tidal channel facies. This interval exhibits the best reservoir quality in the 15/9-19BT2 well; it is predominantly composed of rock type 4, reflecting good reservoir performance throughout the zone.

This integrated workflow illustrates the efficiency of machine learning in automating core interpretation and enhancing subsurface understanding of this mature offshore field.

Introduction

In traditional reservoir evaluation approaches, reservoir properties are often assessed independently, with limited integration between sedimentological and petrophysical data (e.g., Yang et al., 2021; Hou et al., 2023; Hasoon and Farman, 2024). Sedimentological analysis focuses on core description, facies interpretation and architecture, diagenetic features, and pore network characterization, but typically lacks a quantitative assessment of reservoir properties. This process relies heavily on visual observation, manual logging, and thin section analysis, making it time-consuming, resource-intensive, and prone to interpreter bias. On the other hand, petrophysical analysis provides a quantitative evaluation of reservoir properties using well log responses, including gamma ray, density, neutron porosity, and sonic logs. It enables the estimation of lithology, porosity, permeability, fluid saturation, shale volume, and clay content. However, these measurements are indirect and relatively low in resolution, often failing to capture mesoscale to microscale features such as thin beds and diagenetic heterogeneities, features that can critically influence in reservoir quality.

Traditional workflows also tend to underutilize machine learning models that could streamline and enhance interpretation (Hou et al., 2022). The application of supervised and unsupervised algorithms supports pattern recognition, improves consistency, and helps uncover subtle reservoir trends that may be overlooked through conventional methods.

To address these challenges, this study presents an integrated, multi-scale reservoir quality evaluation workflow for the Middle to Late Jurassic reservoir in the Volve Field (Hugin Formation), Norwegian North Sea. The proposed approach integrates geological and petrophysical data across spatial scales and applies machine learning algorithms to improve interpretation from reservoir to pore scale. This workflow enhances characterization efficiency and accuracy, enables a more detailed representation of reservoir heterogeneity, reduces uncertainty, and supports significant time and cost savings in reservoir evaluation.

The Volve Field is located approximately 200 km west of Stavanger, Norway, within block 15/9 in the central part of the North Sea (Fig. 1). It lies 8 km north of the Sleipner Øst structure and is part of the prolific North Sea Jurassic petroleum system. The field was developed within the Middle to Upper Jurassic Hugin Formation, which forms the primary reservoir unit and consists predominantly of shallow marine to marginal marine sandstones. The reservoir is defined by a combination of stratigraphic and structural trapping mechanisms, influenced by syn-depositional faulting and extensional faulting during the Middle Jurassic to Early Cretaceous rifting (Zabanbark, 2012).

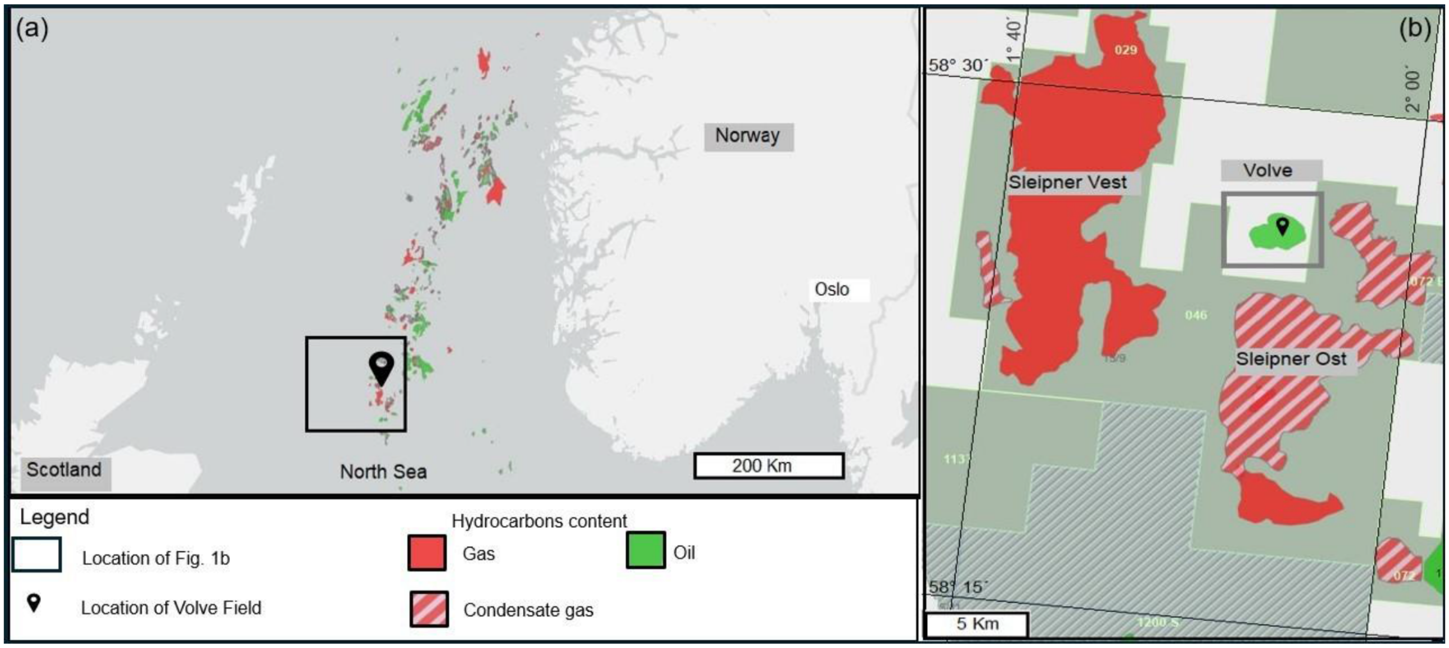

Figure 1—(a) Location of study area, Volve Field, North Sea (adapted from Norwegian Offshore Directorate). (b) Detailed location of the Volve Field, positioned north of the Sleipner Øst structure (adapted from Norwegian Offshore Directorate).

The Hugin Formation, deposited during the Late Bajocian to Oxfordian, represents a complex interplay of depositional environments including tidal flats, estuarine channels, shoreface, and deltaic systems (Otoo and Hodgetts, 2021). These environments give rise to heterogeneous reservoir architecture with variable lithology, grain size, sedimentary structures, and bioturbation intensity. Diagenetic processes such as cementation further influence reservoir quality by modifying porosity and permeability (Jiratitipat, 2020).

Reservoir quality is strongly controlled by lithofacies distribution, diagenetic overprints, and internal heterogeneities, which are not always clearly resolved through conventional petrophysical analysis alone. Therefore, detailed sedimentological characterization integrated with petrophysical based data is essential for accurate reservoir evaluation.

This study focuses on the multi-scale reservoir evaluation of two wells from the Volve Field, Volve 15/9-19A and 15/9-19BT2 (Fig. 1). It aims to integrate sedimentological and petrophysical data, along with machine learning techniques to better understand the reservoir heterogeneity and quality within the Hugin Formation.

Geological and stratigraphic framework

The Volve Field is located approximately 200 km west of Stavanger, Norway, within Block 15/9 in the central part of the North Sea. It lies about 8 km north of the Sleipner Øst structure (Fig. 1). The Theta Vest structure, which defines the Volve Field, is situated between the Loke structure to the east and the northwestern extension of the Sleipner Øst structure. This structure is domal in shape and has undergone complex tectonic deformation. It is located east of a major bounding fault (Jiratitipat, 2020; Saha et al., 2022).

This region experienced two major extensional tectonic phases. The first occurred during the Permian/Triassic transition, initiating rift development, while the second, more dominant phase took place from the Middle Jurassic to Early Cretaceous (Cross, 2022). This later rifting event played a key role in shaping the rift-related architecture of Central Graben and controlling the structural configuration of the Volve Field.

The primary reservoir unit in the Volve Field is the Middle/Upper Jurassic Hugin Formation, which was deposited during the Late Bajocian to Oxfordian stages. It is part of the Vestland Group and comprises shallow-marine and marginal-marine deposits that interfinger with the continental deposits of the underlying Sleipner Formation and the offshore marine shales of the overlying Heather Formation (Partington et al., 1993; Kieft et al., 2011).

The Hugin Formation comprises fine- to coarse-grained sandstone succession with subordinate siltstones and mudstones interbeds. Thickness can exceed 150 m locally, and reservoir quality is highly variable due to lateral and vertical facies changes. Characterized by low gamma-ray responses and high net-to-gross ratios, the Hugin Formation is a key reservoir target in the Central North Sea. The depositional architecture reflects a dynamic interplay of estuarine, tidal flat, shoreface, and deltaic systems, shaped by fluctuating sea levels and variable sediment supply during Middle to Late Jurassic transgressive-regressive cycles.

Data and Methods

This study utilizes publicly available datasets from Equinor's Volve Field, specifically from 15/9-19A and 15/9-19BT2 wells (Equinor). The datasets include high-resolution core images, thin-section microphotographs, petrophysical well logs, core gamma-ray logs, and routine core analysis (RCA) results. They cover a total depth interval of 64 m for 15/9-19A well and 72 m for 15/9-19BT2 well.

A multi-scale reservoir characterization workflow was implemented using the Stratbox Core Explorer® platform (CE; Fig. 2). The workflow begins (Fig. 3) with the integration of core gamma-ray data and the analysis of 133 high-resolution core images to identify lithological variations, sedimentary structures, and bedding contacts. A gamma-ray cut-off value of 3300 c/min was applied to distinguish sandstone from shale and establish preliminary lithological boundaries.

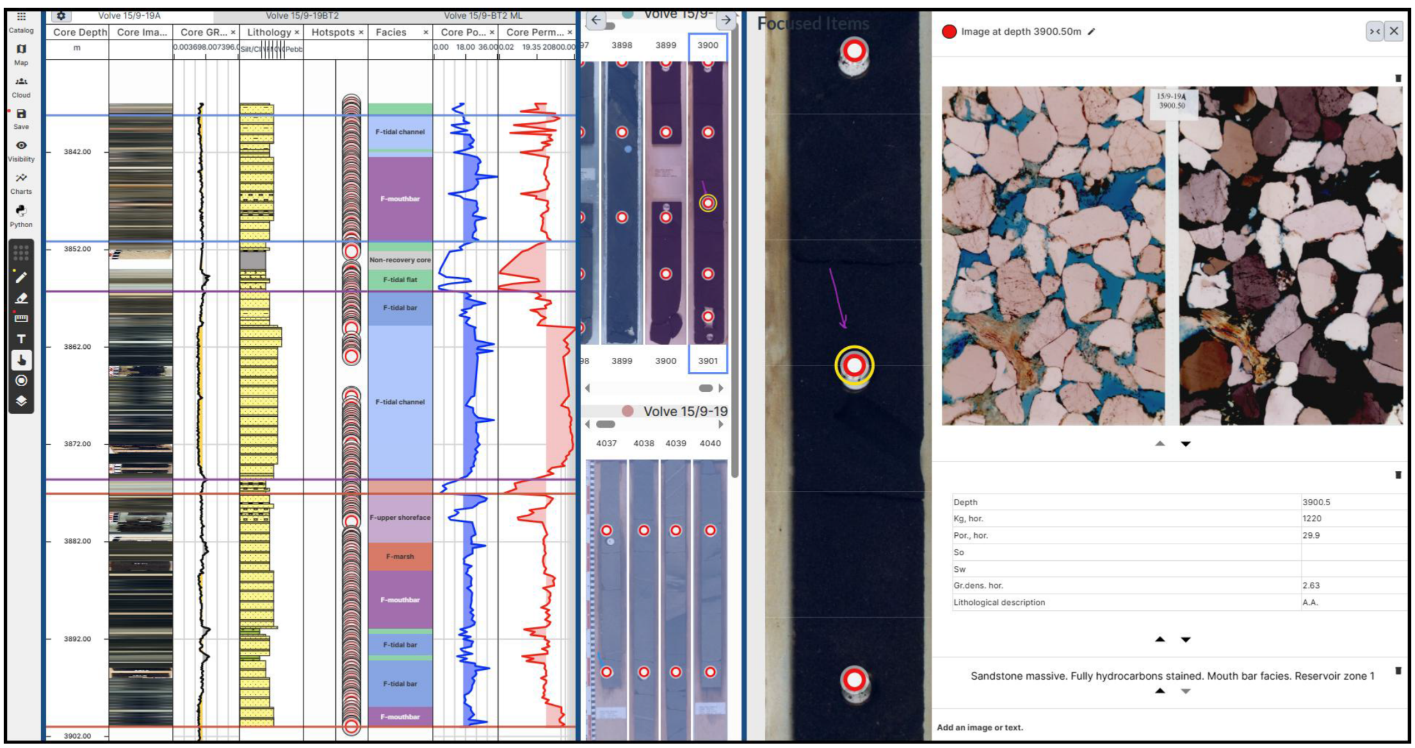

Figure 2—Multi-scale reservoir characterization at core data integration platform.

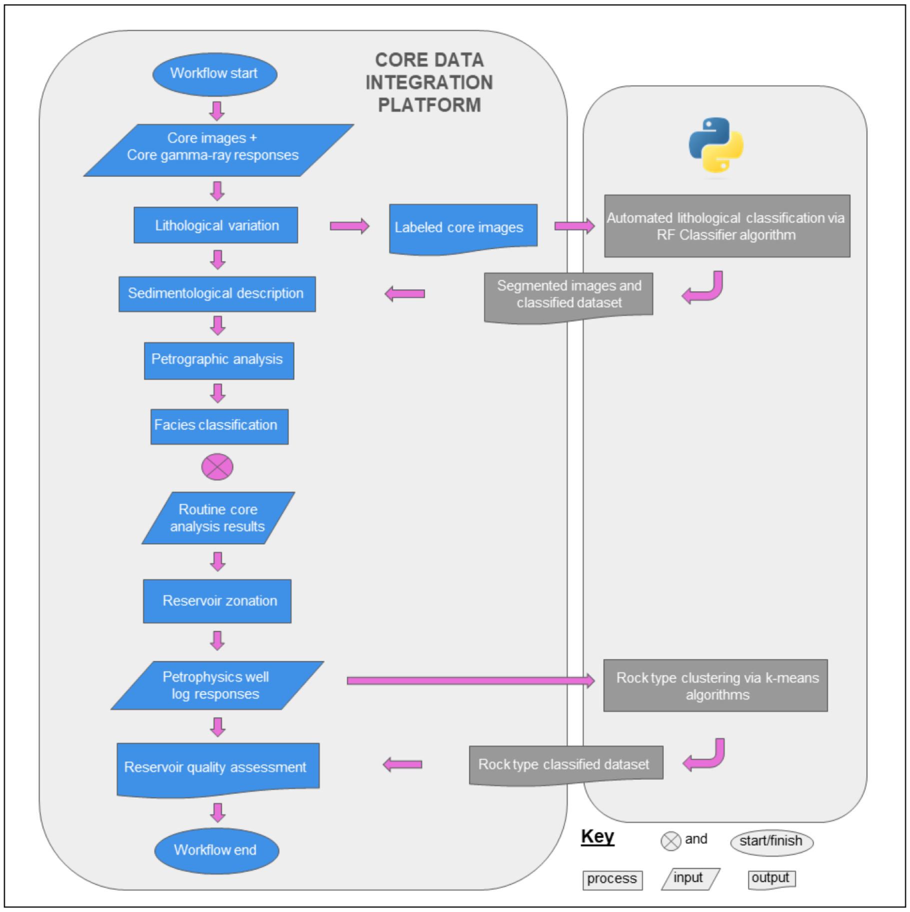

Figure 3—Multi-scale reservoir characterization workflow. It shows the interaction between core data integration platform and Python.

To enhance this interpretation, automated lithological classification was conducted on core images from 15/9-19BT2 well. These images have a resolution of 37.8 pixels/cm and depth coverage of 43 m for the analyzed interval. Each image was depth-registered to the core depth reference system and visually inspected to exclude frames affected by lighting artifacts or scale distortions. The classification workflow began with manual labeling of representative core image segments based on lithological class and dominant color characteristics. Lithological boundaries derived from gamma-ray analysis were used to segment the core images. A total of 34 labeled segments, representing seven lithological classes, were then preprocessed and subjected to feature extraction within a Python environment via the Stratbox Core Explorer Python API (Python API; Fig. 3). In Python, the labeled images were retrieved from this platform, loaded into a variable, and prepared for subsequent processing steps. These steps included: (1) associating colors with feature labels from CE's interpretation tools, (2) preprocessing and segmentation, (3) extract multiscale features (intensity, edges, texture) from the original images, and (4) defining predictors and targets for model training.

Then a Random Forest (RF) model was implemented using the open-source scikit-learn library (Pedregosa et al., 2011). Input features included multiscale intensity, edge, and texture descriptors extracted from core images, complemented by gradient-based features to capture fine scale lithological patterns. RF was trained to recognize visual patterns associated with each lithological class and then applied to predict classifications across the full core image set. The dataset was partitioned into training (80%) and testing (20%) subsets, with a fixed random seed (42) to ensure reproducibility. Default hyperparameters were used, except for n_estimators (number of trees) = 50, max_depth (maximum tree depth) = 20, and max_samples (fraction of data per tree) = 0.6. Classifier performance was evaluated on the held-out test set using precision, recall, and F1-score metrics, with a confusion matrix providing additional insight into classification errors across the lithological classes. The trained model was then applied to the entire core image set, producing continuous depth-based lithological classification and classification data set. These outputs were exported back to CE to complete the sedimentological description of the 15/9-19BT2 well. Automated RF classifications provided the initial lithological framework, which was then refined by petrographic thin-section analysis to capture microscale heterogeneity.

Further refinement was achieved through petrographic analysis of 127 thin-section microphotographs for detailed facies analysis. Petrographic interpretation focused on grains composition, cement content, and pore network structure for each facies. This facies classification was then correlated with RCA results, including core porosity and permeability, to better constrain facies specific reservoir quality.

Finally, a K-means clustering approach was applied to classify rock types based on petrophysical well logs from the Volve dataset within a Python environment via the Python API (Fig. 3). Input variables included permeability (KLOGH_mD), effective porosity (PHIF_v/v), resistivity (RT_ohm.m), gamma ray (GR_GAPI), bulk density (RHOB_g/cm3), sonic (DT_us/ft), and water saturation (SW_v/v), summing up 2361 log responses. Well log curves were depth-matched to core reference and quality-controlled to remove spurious spikes and tool-related gaps. Log curves were selected according to their measuring capabilities, like porosity, and permeability measure the reservoir's storage and deliverability capacity, whereas saturation of water indicates the presence of hydrocarbons in the reservoir of interest. GR_GAPI and RHOB_g/cm3 logs measure the rock type based on lithology and RT_ohm.m reflects hydrocarbons saturation. In Python, the well log dataset was retrieved from this platform, loaded into a variable, and prepared for subsequent processing steps. After data quality control, the variables were transformed into logarithmic scale. Exploratory data analysis (EDA) was then performed to evaluate variable distributions and interrelationships. A pair plot was generated to visualize the distribution of each variable across wells and also crossplot the variables against each other. Additionally, a correlation matrix quantified the strength and direction of variable interrelationships.

Principal Component Analysis (PCA) was applied to reduce dimensionality and select the most informative features for clustering, using default hyperparameters. K-means clustering, implemented with the open-source scikit-learn library (Pedregosa et al., 2011), was then applied to classify each depth interval into distinct rock types. The optimal number of clusters was determined using the elbow and silhouette methods, in conjunction with validation against geological interpretations. K-means hyperparameters were k (number of clusters) = 6, init (initialization)= k-means++, and max_iter = default. The model generated a rock type classification dataset for the study wells, which was exported back to CE to complete the multi-scale reservoir characterization workflow. This unsupervised machine learning approach enabled objective, data-driven reservoir zonation, and quality assessment across the study wells.

Results and conclusions

Sedimentological description

The Hugin Formation in 15/9-19A and 15/9-19BT2 wells is characterized by a sandstone-dominated succession, with subordinate occurrences of siltstone, silty claystone, and claystone.

In Volve 15/9-19A well, this unit is predominantly composed of fine to coarse-grained sandstones exhibiting planar cross-bedding and low-angle cross-bedding. These sedimentary structures are commonly diffuse and, in some intervals, are completely obscured due to pervasive hydrocarbon impregnation. A gradual decrease in hydrocarbon saturation is observed toward the upper part of the well, specifically between 3837 and 3860 m. Within this interval, bioturbated clayey sandstones are frequently distributed between 3837 and 3842 m. Additionally, subordinate thin beds (0.5-1.5 m of thickness) of calcareous sandstones and laminated dolomitic sandstones are identified throughout this well, specially into the lower (3875.6-3901 m) and upper (3838.19-3856.28 m) sections. These carbonate-rich intervals show no visible signs of hydrocarbon impregnation.

In 15/9-19BT2 well, the Hugin Formation displays greater lithological variability, as identified through automated lithological classification. This approach enhanced the consistency and reproducibility of the core description, providing a robust lithological framework to support sedimentological interpretation. A more detailed description of this well is presented in the following subsection.

Automated lithological classification. Automated classification was applied to 40 high-resolution core images from 15/9-19BT2 well, covering a total depth interval of 43 m. This well is primarily composed of very fine to coarse-grained sandstones exhibiting low-angle cross-bedding and planar cross-bedding. Using a RF classifier model trained on labeled core image segments, the algorithm successfully differentiated previously established categories into seven main lithology classes: stained sandstones, laminated sandstones, clean sandstones, carbonaceous laminated sandstones, carbonaceous bioturbated sandstones, calcareous sandstones and cross-laminated sandstones (Fig. 4). Confusion matrix indicates strong classification across most lithologies, with limited overlap between visually similar classes. Visual inspection shows that misclassifications primarily occurred in thinly interbedded zones, where textural contrasts were less distinct, or fragmented intervals. The RF classifier achieved precision, recall and f1-score values between 90 to100%.

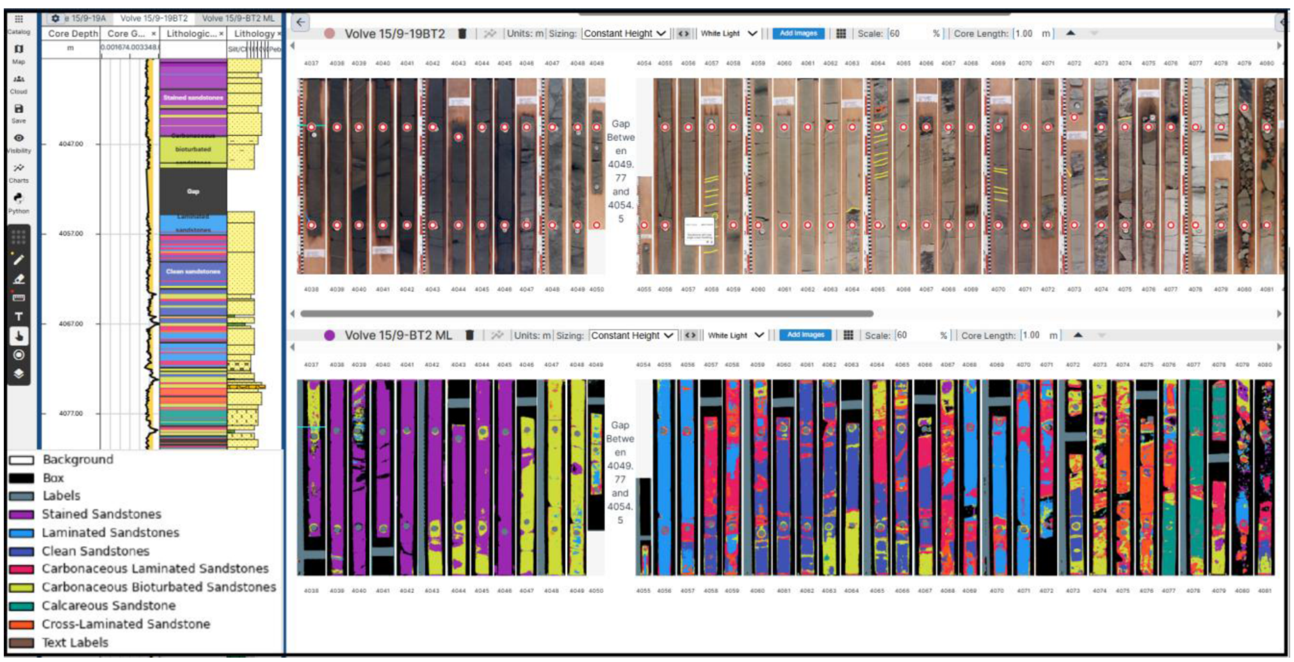

Figure 4—Automated lithological classification using random forest classifier. It shows the seven main lithological classes differentiated by the random forest algorithm.

Into the lower section of 15/9-19BT2 well (4079.15-4109 m), carbonaceous bioturbated, carbonaceous laminated sandstones, clean sandstone and laminate sandstone dominates. Toward the top of this section calcareous sandstones appear. The middle section (4054.50–4079.15 m) is characterized by greater lithological variability, including calcareous, cross-laminated, and carbonaceous bioturbated sandstones, particularly between 4072 and 4079.15 m, whereas laminated sandstones, carbonaceous laminated, and clean sandstones are more prevalent from 4054.50 to 4072 m. The upper section (4037-4054.50 m) is distinguished by the predominance of carbonaceous bioturbated and hydrocarbon-stained sandstones. Particularly stained sandstones are representing from 4037 to 4046.50 m (Fig. 4).

The automated lithological classification applied to 15/9-19BT2 well not only provided a detailed and reproducible lithological framework but also demonstrated clear operational advantages over traditional core description. Compared with the traditional sedimentological description of well 15/9-19A, the automated workflow significantly reduced the time required to analyse long-cored intervals, achieving time savings of approximately 30–40%. This gain in efficiency, combined with high classification accuracy and consistent reproducibility, underscores the potential of automated approaches to enhance sedimentological interpretation and to accelerate reservoir characterization workflows.

Facies and petrophysical integration

Facies analysis led to the identification of eight distinct depositional facies: tidal flat, tidal channel, tidal bar, mouth bar, marsh, upper shoreface, lower shoreface, and offshore. These facies represent a spectrum of depositional environments, ranging from marginal marine to fully marine settings, and exhibit distinct petrophysical characteristics that influence porosity and permeability distributions.

Tidal flat facies comprise interlaminated clayey sandstones, siltstones, and claystones, along with calcareous and dolomitic sandstones. These intervals are characterized by lenticular to planar laminations and moderate bioturbation. Porosity and permeability values are low, primarily due to the high clay and silt content and pervasive carbonate cementation.

Tidal channel facies are composed of medium to coarse-grained sandstones with planar cross-bedding, commonly hydrocarbon-stained. These deposits typically preserve high porosity and permeability through well-connected poral network, though mud drapes and localized cementation can reduce reservoir quality.

Tidal bar facies include fine to medium-grained sandstones to carbonaceous sandstones, with planar to low-angle cross-bedding and common bioturbation. These facies generally show moderate to high porosity and permeability, but localized calcareous cementation may diminish these values.

Mouth bar facies consist of fine to medium-grained sandstones to carbonaceous sandstones. Structures range from massive to low-angle cross-bedding, with limited bioturbation. These facies exhibit moderate to high porosity and permeability, however, features such as localized levels of calcareous and dolomitic sandstones can reduce these values.

Marsh facies are composed of massive, fine-grained sandstones with moderate kaolinite cement content. These intervals are associated with low porosity and permeability, primarily due to clay-rich matrices and diagenetic kaolinite cementation.

Upper shoreface facies include well-sorted, fine-grained laminated sandstones to calcareous sandstones. These intervals show moderate porosity and permeability, although cementation may locally reduce pore connectivity.

The lower shoreface facies consist of very fine to fine-grained, planar-laminated sandstones to calcareous sandstones. Porosity and permeability are generally low to moderate, with cementation further reducing poral network effectiveness. Finally, the offshore facies are dominated by fine-grained, laminated sandstones to claystones. These are typically very thin intervals, with very low porosity and permeability.

The integration of facies and petrophysical data highlights the strong control of depositional environment and diagenetic processes on reservoir quality within the studied succession. High-energy depositional settings, such as tidal channels, tidal bars, and mouth bars, provide the most favourable reservoir intervals. These facies are dominated by medium to fine-grained sandstones with well-preserved porosity and limited cementation, resulting in high permeability and effective pore connectivity. In contrast, low-energy settings including tidal flats, marshes, and offshore deposits show consistently poor reservoir quality. Their high clay content, fine-grained textures, and pervasive diagenetic cementation significantly reduce porosity and permeability. Upper and lower shoreface settings display moderate reservoir quality, with localized cementation exerting a strong influence on pore network continuity.

Overall, the results emphasize that reservoir heterogeneity is primarily governed by the interplay between depositional facies, diagenetic overprint and clay content. The distribution of tidal channel, tidal bar and mouth bar facies within the stratigraphic framework therefore represents the primary control on hydrocarbon storage and flow, while low-permeability facies act as internal barriers.

Rock type

Rock type identification was performed using K-means clustering to classify well log data into discrete reservoir quality classes. Following data quality control and preprocessing, EDA was conducted to assess variable distributions and correlations. A pair plot analysis and correlation matrix visualization identified KLOGH_mD, RT_ohm.m, PHIF_v/v, GR_GAPI, and RHOB_g/cm[3] as the most relevant logs for clustering, based on the good correlation between these variables (Fig. 5).

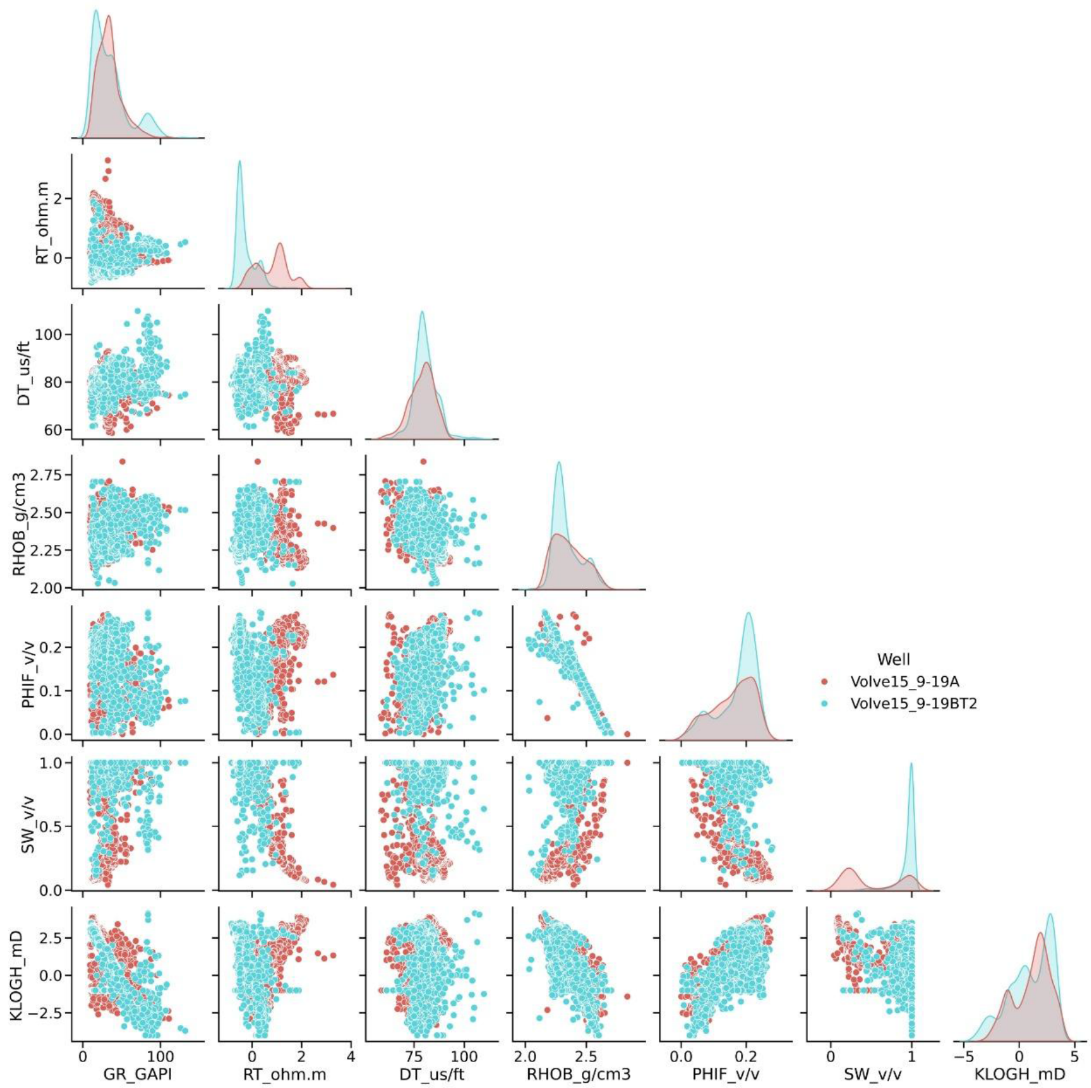

Figure 5—Pair plot showing cross plots among GR_GAPI, RT_ohm.m, DT_us/ft, RHOB_g/cm³, PHIF_v/v, SW_v/v and KLOGH_mD to visualize distribution of each log for 15/9-19A and 15/9-19BT2 wells.

Dimensionality reduction was achieved through PCA. The first three components explain 81.0%, 15.8%, and 1.25% of the total variance, respectively. KLOGH_mD most significantly contributes to the first principal component; RT_ohm.m dominates the second, and PHIF_v/v has the strongest influence on the third. These results along with EDA guided the selection of KLOGH_mD, RT_ohm.m, PHIF_v/v, GR_GAPI, and RHOB_g/cm3 log as the input features for clustering.

The optimal number of clusters (k) was determined using the elbow and silhouette methods, which suggested a range between 4 and 6 clusters. Silhouette coefficients across this range varied narrowly between 0.49 and 0.50. A final k value of 6 was selected, supported by both statistical evaluation and alignment with sedimentological interpretation. Once defined, cluster analysis was performed, generating the output curve "Rock type".

The clustering results delineate six distinct rock types reflecting varying levels of reservoir quality. Mean values of the selected petrophysical logs for each rock type are summarized in Table 1. The classification follows an ascending order of reservoir quality, from very poor (rock type 1) to excellent (rock type 6). Rock types 4 and 5 represent good-quality intervals but differ in fluid content. Rock type 4 exhibits low resistivity (0.38 ohm.m) and high-water saturation (0.95 v/v), suggesting water-bearing zones. In contrast, rock type 5 shows high resistivity (21.60 ohm.m) and low-water saturation (0.27 v/v), indicating hydrocarbon saturated intervals. On the other hand, rock type 6 represents excellent reservoir quality, shows the highest resistivity (52.69 ohm.m) and the lowest water saturation (0.27 v/v) values, suggesting hydrocarbon-bearing zones.

Table 1—Showing reservoir quality based on Rock Type classification along with their respective color code. Each rock type shows the mean values of the selected petrophysical well logs.

Rock Type	Reservoir Quality	Data Points	Rock Type Mean Values						
			GR_GAPI	RT_ohm.m	DT_us/ft	RHOB_g/cm3	KLOGH_mD	PHIF_v/v	SW_v/v
1	Very Bad	193	72.622	2.259	78.867	2.539	0.005	0.061	0.995
2	Bad	421	43.227	4.612	77.592	2.469	0.208	0.11	0.855
3	Moderate	534	39.952	1.203	81.334	2.355	8.759	0.167	0.883
4	Good	632	20.08	0.386	78.952	2.281	734.505	0.205	0.954
5	Good	371	32.388	21.605	82.205	2.29	106.366	0.197	0.267
6	Excellent	210	18.795	52.694	81.665	2.22	2231.973	0.218	0.196

The clustering results were validated through comparison with facies analysis and RCA data, including porosity, permeability, water saturation, and oil saturation. Rock types 4 to 6 show the best correlation with favorable facies and RCA parameters. In the 15/9-19A well, most of the interval corresponds to rock types 5 and 6, indicating good to excellent reservoir quality, with fewer sections falling into lower-quality categories. The high proportion of rock type 5 and 6 in this well suggests significant hydrocarbons saturation within this reservoir, also supported by sedimentological analysis. In contrast, the 15/9-19BT2 well is predominantly composed of rock types 3 and 4, representing moderate to good quality, with limited intervals of poorer-quality rock types. The high proportion of rock type 4 in this well suggests significant water saturation within the reservoir zone.

Reservoir Zonation

Multi-scale reservoir characterization within the Hugin Formation enabled identification of three distinct reservoir quality zones in the Volve 15/9-19A well (Fig. 6), and two zones in the 15/9-19BT2 well (Fig. 7). These zones are separated by barriers represented by lithological and diagenetic heterogeneities, primarily corresponding to poor reservoir quality (rock type 2).

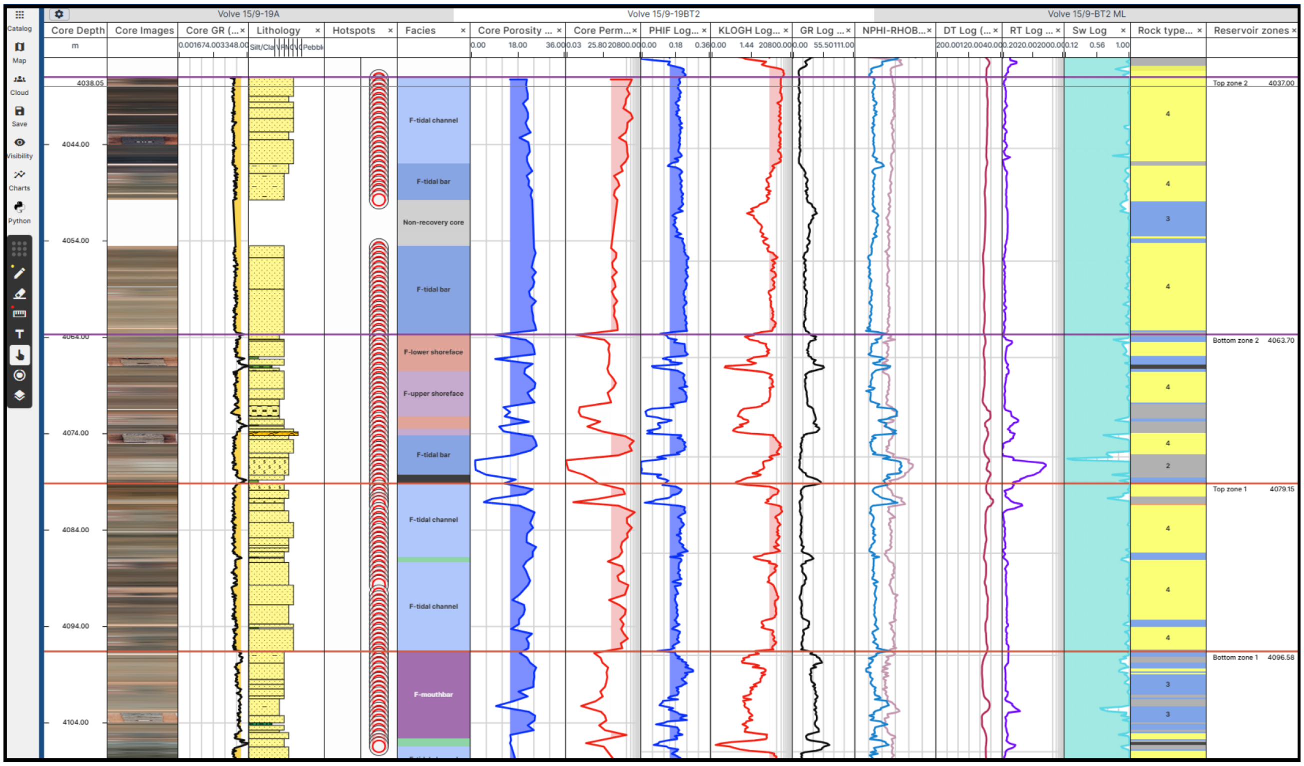

Figure 6—Reservoir quality zones in the Volve 15/9-19A well.

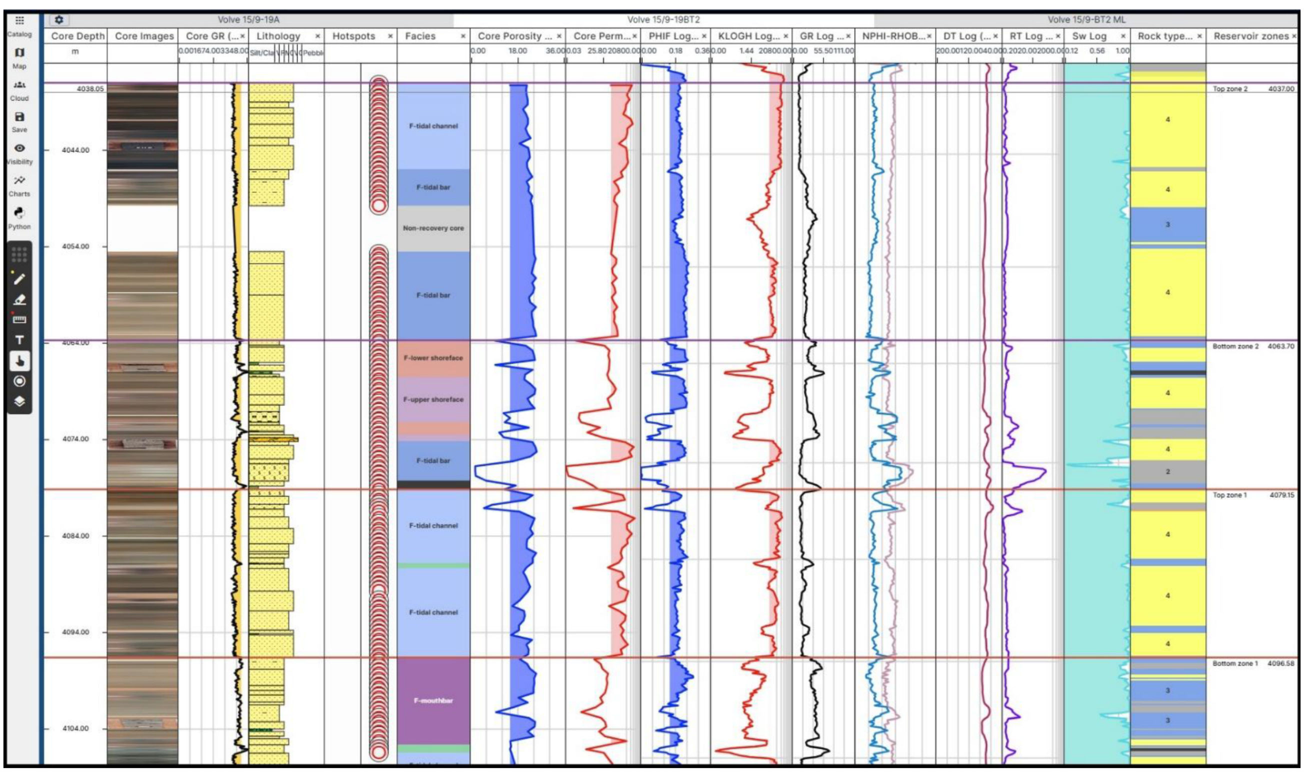

Figure 7—Reservoir quality zones in the Volve 15/9-19BT2 well.

Reservoir zone one (23.74 m thick) on the Volve 15/9-19A well, is dominated by mouth bar, tidal bar, and upper shoreface facies. Porosity spans 15–31.3% and permeability varies between 40-2780 mD. This interval is primarily associated with good reservoir quality (rock type 5), with localized sections of excellent quality (rock type 6). However, the presence of heterogeneities, such as tidal flat and marsh facies, results in localized reductions in porosity (8.6–23.2%) and permeability (0.58–218 mD). These heterogeneities correspond to poor to moderate reservoir quality.

Reservoir zone two (19.32 m thick) is dominated by tidal channel facies, with minor contribution from tidal bar facies. It exhibits the highest reservoir quality in the 15/9-19A well, with porosity ranging from 14.7-33.8% and permeability varies between 15-20800 mD. This zone is largely associated with excellent reservoir quality (rock type 6), with subordinate intervals of good quality (rock type 5).

Reservoir zone three (12.96 m thick), is composed mainly of mouth bar and tidal channel facies, with moderate petrophysical properties (porosity: 12.7-36%, permeability: 14.5-1130 mD). It is predominantly characterized by good reservoir quality (rock type 5), with minor sections exhibiting excellent quality (rock type 6). However, the presence of lithological heterogeneities (represented by tidal flat facies) and diagenetic overprint within the mouth bar facies leads to localized reductions in porosity (8.7–22%) and permeability (0.15–61 mD), corresponding to poor reservoir quality.

In the Volve 15/9-19BT2 well, two reservoir quality zones were identified within the Hugin Formation. Reservoir zone one (17.43 m thick) is dominated by tidal channel facies. Porosity ranges from 12.8% to 24.9%, and permeability varies between 59.3 and 7360 mD. This interval is primarily characterized by good reservoir quality (rock type 4), with minor sections exhibiting moderate quality (rock type 3). Limited diagenetic heterogeneities are observed near the top of the zone, locally reducing reservoir quality.

Reservoir zone two (26.70 m thick), comprises a mix of tidal bar and tidal channel facies. This interval exhibits the best reservoir quality in the 15/9-19BT2 well, with porosity consistently above 15% (ranging from 18.6 to 24.9%) and permeability exceeding 100 mD (106 to 5110 mD). It is predominantly composed of rock type 4, reflecting good reservoir performance throughout the zone.

Conclusions

This study demonstrates the value of integrating machine learning with sedimentological and petrophysical analyses, to achieve a multi-scale reservoir characterization of the Hugin Formation in the Volve Field. Automated lithological classification using Random Forest algorithms significantly improved the efficiency, and reproducibility of core description, reducing interpretation time by up to 40% while maintaining high accuracy. The integration of facies and petrophysical data provided robust insights into the controls of depositional environment and diagenetic processes on porosity and permeability distributions. As a result, tidal channel, tidal bar, and mouth bar facies emerge as the most favorable reservoir intervals.

Rock typing through K-means clustering of well log data enabled the delineation of six discrete reservoir quality classes, validated against facies and routine core analysis. This approach revealed contrasting reservoir performances between the two studied wells: 15/9-19A contains a higher proportion of hydrocarbon-saturated, good to excellent quality intervals (rock types 5–6), whereas 15/9-19BT2 is dominated by water-saturated, moderate to good quality rock types (3–4). The integration of facies analysis, petrophysical evaluation, and unsupervised clustering enabled the identification of discrete reservoir zones and the recognition of heterogeneities that control storage and flow within the Hugin Formation. Three reservoir quality zones were delineated in 15/9-19A well and two in 15/9-19BT2 well, separated by lithological and diagenetic heterogeneities. These results highlight the value of multi-scale integration for capturing reservoir complexity and reducing uncertainty in mature offshore fields

Overall, the proposed workflow illustrates how combining supervised and unsupervised machine learning and traditional geological methods can enhance subsurface evaluation. Beyond the Volve Field, this methodology offers a scalable and transferable framework to other hydrocarbon reservoirs, supporting more efficient field development, optimized reservoir management, and reduced uncertainty in exploration and production decision-making.

References

Cross, L. C. 2022. *Tectonostratigraphic evolution of the North Sea rift basin*. PhD dissertation, Colorado School of Mines, Colorado, USA.

Equinor. *Equinor Data Sharing Volve Field Dataset*. Available online at https://data.equinor.com/dataset/Volve (accessed on 11 September 2020).

Hasoon, S. and Farman, G. 2024. A comprehensive review for integrating petrophysical properties, rock typing, and geological modeling for enhanced reservoir characterization. *Journal of Engineering* **30** (10): 69–101. https://doi.org/10.31026/j.eng.2024.10.05.

Hou, J., Zhao, L., Zeng, X. et al 2022. Characterization and evaluation of carbonate reservoir pore structure based on machine learning. *Energies* **15**: 7126. https://doi.org/10.3390/en15197126.

Hou, J., Zhao, L., Zhao, W. et al 2023. Evaluation of pore-throat structures of carbonate reservoirs based on petrophysical facies division. *Frontiers in Earth Science* **11**: 1164751. https://doi.org/10.3389/feart.2023.1164751.

Jiratitipat, T. 2020. Mapping of Fault System Related to Salt Movement in Volve Field, Offshore Norway, North Sea. *Bulletin of Earth Sciences of Thailand* **12**: 25–36.

Kieft, R. L., Jackson, C. A. L., Hampson, G. J. and Larsen, E. 2010. Sedimentology and sequence stratigraphy of the Hugin Formation, Quadrant 15, Norwegian sector, South Viking Graben. In *Petroleum Geology: From Mature Basins to New Frontiers – Proceedings of the 7th Petroleum Geology Conference*, ed. B.A. Vining and S.C. Pickering, Chap. 7, 157–176. London, UK: Queen Elizabeth II Conference Centre.

Norwegian Offshore Directorate. *Map of Volve Field Location*. Available online at https://www.sodir.no/en (accessed on 4 August 2025).

Otoo, D. and Hodgetts, D. 2021. Porosity and permeability prediction through forward stratigraphic simulations using GPM. *Geoscientific Model Development* **14** (4): 2075–2095. https://doi.org/10.5194/gmd-14-2075-2021.

Partington, M. A., Mitchener, B. C., Milton, N. J. et al 1993. Genetic sequence stratigraphy for the North Sea Late Jurassic and Early Cretaceous: Distribution and prediction of Kimmeridgian–Late Ryazanian reservoirs in the North Sea and adjacent areas. *Geological Society, London, Petroleum Geology Conference Series* **4** (1): 347–370. London, UK: The Geological Society of London.

Pedregosa, F., Varoquaux, G., Gramfort, A. et al 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* **12**: 2825–2830.

Saha, S., Vishal, V., Mahanta, B., Pradhan, S.P. et al 2022. Geomechanical model construction to resolve field stress profile and reservoir rock properties of Jurassic Hugin Formation, Volve Field, North Sea. *Geomechanics and Geophysics for Geo-Energy and Geo-Resources* **8**: 68. https://doi.org/10.1007/s40948-022-00359-5.

Yang, J., Wang, E. and Ji, Y. 2021. Diagenetic facies and reservoir porosity evaluation of deep high-quality clastic reservoirs: A case study of the Paleogene Shahejie Formation, Nanpu Sag, Bohai Bay Basin, China. *Energy Exploration & Exploitation* **39** (4): 1097–1122. https://doi.org/10.1177/0144598721998504

Zabanbark, A. 2012. Geological structure and petroleum resource potential of the North Sea Basin. *Oceanology* **52** (4): 513–525. https://doi.org/10.1134/S0001437012030125.